

Souvik Banerjee

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Experience

Amla Labs

Founder · South Park Commons · Pre-seed

Oakland, CA

Mar 2026-Present

- Building the security layer for AI agents: capability-based authorization, microVM isolation, and cryptographic delegation chains that bind authority to individual transactions rather than long-lived identities.
- Shipped `amla-sandbox`, a 13MB WebAssembly runtime giving agents capability-scoped POSIX execution with deterministic replay; cuts LLM context overhead by ~98% versus raw tool-call patterns.
- Shipped Kalahari, a TypeScript microVM sandbox running locally on KVM and Hypervisor.framework with millisecond zygote-based spawning and drop-in compatibility for E2B, Daytona, and ComputeSDK.

Tudor Investment Corporation

Quantitative Developer: Macro Pipeline

Oakland, CA

Nov 2024 - Jun 2026

- Quantitative research and development supporting low-latency trading strategies and systems.
- Architected and deployed a distributed system to transmit predictive signals across exchanges, credited with unlocking \$10M+ in annual P&L through cross-venue strategy execution.
- Engineered a custom market data ingestion pipeline, streaming compressed packet capture files directly from AWS S3 and eliminating intermediate decompression; reduced annual cloud processing and storage costs by approximately \$250,000.

Apple

Senior Software Engineer: Core OS Kernel Technologies

Cupertino, CA

Sep 2019 - Nov 2024

- Directed engineering effort to integrate ARMv9 Memory Tagging Extensions (MTE) into iOS secure kernel and firmware.
- Developed device driver infrastructure for the Apple N1 wireless chip, shipped in [iPhone Air](#).
- Led teams building novel confidential compute operating systems, shipped in iOS 17 to >1B devices.
- Architected and delivered new features for third party device drivers on iPad, adding support for custom USB and Thunderbolt devices. Presented at [WWDC 2022](#).
- Designed and implemented core security and stability enhancements for Apple operating systems.

Software Engineer Intern: Core Kernel

May 2018 - Aug 2018

University of Texas at Austin

Undergraduate Researcher

Austin, TX

Aug 2017 - Aug 2019

- Prototyped new blockchain architectures with a focus on network and storage performance.
- Publications:
 - Ponnappalli, Soujanya, Aashaka Shah, **Souvik Banerjee**, Dahlia Malkhi, Amy Tai, Vijay Chidambaram, and Michael Wei. "RainBlock: Faster Transaction Processing in Public Blockchains." In 2021 USENIX Annual Technical Conference (USENIX ATC 21).

Teaching Assistant

Aug 2016 - Dec 2017

- Taught CS 439H Operating Systems Honors, CS 429H Computer Architecture Honors, CS 311 Discrete Mathematics.

Point72: Cubist Systematic Strategies

Research Analyst Intern: FX Options Trading

New York, NY

Dec 2018

Citadel Securities

Software Engineer Intern: Low Latency

Chicago, IL

Sep 2018 - Nov 2018

Facebook

Software Engineer Intern: WhatsApp iPhone

Menlo Park, CA

May 2017 - Aug 2017

Software Engineer Intern: WhatsApp VoIP

May 2016 - Aug 2016

Education

University of Texas at Austin

B.S. Computer Science Honors, Mathematics

May 2019

GPA: 3.989

- **Turing Scholar, Distinguished College Scholar, University Honors**
- Honors Thesis: [Implementing Stateless Clients in Ethereum](#)